Homework 2

Due : Wednesday, April 29 at 11 : 59 pm

Honor Code

Write and sign the following statement:

> "I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

1 Sum of Residuals

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

We have computed the solution w^* to the least squares regression problem:

$$w^* = \arg\min_w \frac{1}{2} \sum_{n=1}^{N} \left(y_n - x_n^\top w \right)^2.$$

The residual vector is then defined as:

$$r = y - Xw^*$$

(a) Use the normal equation to show that: $X^\top r = 0$.

(b) Use the presence of a bias term (encoded in the column of 1s in X) to show that the sum of the residuals is 0.

(c) Suppose we add a point (x_j, y_j) to our dataset with a very large $|y_j|$ but x_j is close to the mean of x. Predict qualitatively how w will change and explain why least squares is sensitive to such a point. How can you modify the loss to make the model more robust to outliers?

2 Weighted Linear Regression

Consider a dataset $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ with $x_n \in \mathbb{R}^{D+1}$ and $y_n \in \mathbb{R}$. Here we assume that X already contains a column of 1s. We model the relationship between x and y using linear regression with parameters $w \in \mathbb{R}^{D+1}$:

$$\hat{y} = x^\top w.$$

Suppose we have nonnegative weights $\{\alpha_n\}_{n=1}^N \geq 0$ for each observation and our goal is to minimize the weighted least squares objective:

$$L(w) \;=\; \sum_{i=1}^{n} \alpha_i (y_i - x_i^\top w)^2.$$

In this problem, we will derive the w^* that minimizes this loss.

(a) Write the weighted least-squares loss function $L(w)$ in matrix form using the diagonal matrix:

$$A = \text{diag}(\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^{N \times N}.$$

Your answer should not contain any summations.

(b) Compute the gradient of the loss and express your answer in matrix form (no summations). You can do this by working with the summation form and taking partial derivatives or by using the following gradient identities:

$$\nabla_w \left(u^\top w \right) = u$$
$$\nabla_w \left(w^\top A w \right) = (A + A^\top) w$$

(c) Set the gradient equal to zero, and solve for the closed-form solution for w^* that minimizes the weighted least squares objective. in terms of X, y, and A, assuming $X^\top A X$ is invertible.

(d) Under what condition is $X^\top A X$ guaranteed to be invertible? Can we derive a simpler condition when $\alpha_i > 0$?

3 ℓ_1 - Regularized Least Squares (Lasso)

Consider a data matrix $X \in \mathbb{R}^{N\times(D+1)}$ with rows $x_n^\top$, a target vector $t \in \mathbb{R}^N$, and a parameter vector $w \in \mathbb{R}^{D+1}$. For a regularization hyperparameter $\lambda \geq 0$, define the lasso objective:

$$E(w) \;=\; \|Xw - t\|_2^2 \;+\; \lambda\|w\|_1,$$

where $\|w\|_1 = \sum_j |w_j|$.

(a) Write $E(w)$ in expanded quadratic form $w^\top Aw - 2b^\top w + c + \lambda\|w\|_1$ by identifying A, b, and c in terms of X and t.

(b) Compute the piecewise derivative of $E(w)$ with respect to w. If any, identify the point(s) where the gradient does not exist.

(c) Explain why a closed-form solution does not exist for linear regression with Lasso regularization.

(d) Assume $w_j > 0$ for all j. Find the optimal parameter vector, w^*.

4 Regression with Maximum Likelihood Estimation

Suppose we have a regression model with parameters $w \in \mathbb{R}^D$ and observations (x_n, t_n) for $n = 1, \ldots, N$. We assume that the observed targets are generated as

$$t_n = x_n^\top w + \epsilon_n,$$

where the noise terms ϵ_n are i.i.d. from a Laplace distribution:

$$p(\epsilon) = \frac{1}{2b}\exp\Big(-\frac{|\epsilon|}{b}\Big)$$

(a) Write down the log-likelihood $\log p(D|w)$ for the parameters w given the dataset $D = \{(x_n, t_n)\}_{n=1}^N$.

(b) Now, place an independent Gaussian prior on the coefficients w. Write down the expression for $p(w)$ and compute log-prior, $\log p(w)$.

(c) Write the Maximum a Posteriori (MAP) objective and show that the MAP estimate for the coefficients w is equivalent to the solution to the least absolute (L1 residual loss) regression problem with ℓ_2-regularization, i.e. minimizing

$$\sum_{n=1}^{N} |t_n - x_n^\top w| \;+\; \lambda\|w\|_2^2$$

I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted.



1) a)

To show that $X^T r = 0$, we will use the definition of the residual vector and the optimality condition of the Ordinary Least Squares solution.

$$L(w) = \frac{1}{2}\sum_{n=1}^{N}(y_n - x_n^T w)^2 = \frac{1}{2}\|y - Xw\|^2$$

$$\nabla_w L(w) = \nabla_w \left(\frac{1}{2}(y - Xw)^T(y - Xw)\right)$$

$$\nabla_w L(w) = -X^T(y - Xw)$$

$$-X^T(y - Xw^*) = 0$$

$X^T X w^* = X^T y$ is known as the Normal Equation

$$r = (y - Xw^*)$$

$$X^T(y - Xw^*) = 0$$

$$X^T r = 0$$

By the property of the least squares solution, the residual vector r is orthogonal to the column space of X. Thus, $X^T r = 0$ is proven.

b)

Show that $\sum_{n=1}^{N} r_n = 0$ given that X contains a column of 1s.

$$X^T r = 0$$

Because of X contains a column of 1s.

$$1^T r = 0$$

$$1^T r = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix} = (1 \cdot r_1) + (1 \cdot r_2) + \dots + (1 \cdot r_N)$$

The summation simplifies to

$$\sum_{n=1}^{N} r_n = 0$$

c)

Qualitative Prediction;

If a point has a very large $|y_j|$ but x_j is near the mean, the intercept will shift significantly toward y_j. The entire regression line will move vertically to try and catch

this distant point.

Sensitivity of Least Squares:
Least squares uses a quadratic penalty. A large outlier creates a massive loss value. To minimize the total loss, the model is forced to prioritize reducing this one large error, even if it makes the predictions for all other points worse.

Robust Modification:
L1 Loss Mean Absolute Error use $|y_n - \hat{y}_n|$. This penalizes errors linearly, making the model ignore extreme outliers.

---

2)
a)
Let the residual vector be $r = y - Xw$
$y \in \mathbb{R}^N$ is the target vector
$X \in \mathbb{R}^{N \times (D+1)}$ is the feature matrix
$w \in \mathbb{R}^{D+1}$ is the weight vector

$$A = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & & \\ 0 & 0 & \cdots & \alpha_n \end{bmatrix}$$

$$L(w) = \sum_{n=1}^{n} a_i r_i^2 \qquad L(w) = r^T A r$$

$$L(w) = (y - Xw)^T A (y - Xw)$$

b)

$$L(w) = (y - Xw)^T A (y - Xw)$$

$$L(w) = (y^T - w^T X^T) A (y - Xw)$$

$$L(w) = y^T A y - y^T A X w - w^T X^T A y + w^T X^T A X w$$

Since $y^T A X w$ is a scalar and A is diagonal

$$(y^T A X w)^T = w^T X^T A^T y \qquad A = A^T$$

$$L(w) = y^T A y - 2 w^T X^T A y + w^T (X^T A X) w$$

$$\nabla_w (y^T A y) = 0$$

$$\nabla_w (-2 w^T X^T A y) = -2 X^T A y$$

$$\nabla_w (w^T X^T A X w) = 2 X^T A X w$$

$$\nabla_w L(w) = -2 X^T A y + 2 X^T A X w$$

$$\nabla_w L(w) = -2 X^T A (y - Xw)$$

$$\nabla_w L(w) = -2 X^T A r \quad (r = y - Xw)$$

c)

$$-2 X^T A (y - X w^*) = 0$$

$$-2 X^T A y + 2 X^T A X w^* = 0$$

$$X^T A X w^* = X^T A y$$

$$(X^TAX)^{-1}(X^TAX)w^* = (X^TAX)^{-1}X^TAy$$

$$w^* = (X^TAX)^{-1}X^TAy$$

d)

X^TAX is invertible if and only if:

-The diagonal Matrix A must have enough non-zero α_i values to support the rank.

-The matrix X must have a rank of D+1.

When $\alpha_i > 0$ for all i, X^TAX is invertible if and only if:

- X has full column rank.

---

3)

a)

$$||Xw - t||_2^2 = (Xw - t)^T(Xw - t)$$

$$(Xw - t)^T(Xw - t) = (w^TX^T - t^T)(Xw - t)$$

$$= w^TX^TXw - w^TX^Tt - t^TXw + t^Tt$$

w^TX^Tt and t^TXw are scalers

$$(w^TX^Tt)^T = t^TXw$$

$$w^TX^Tt + t^TXw = 2t^TXw = 2(X^Tt)^Tw$$

$$w^TAw - 2b^Tw + c = w^T(X^TX)w - 2(X^Tt)^Tw + t^Tt$$

$$A = X^TX \quad b = X^Tt \quad c = t^Tt$$

b)

$$E(w) = w^T A w - 2b^T w + c + \lambda \sum_j |w_j|$$

Derivative of the Quadratic Part

$$\nabla_w (w^T A w - 2b^T w + c) = 2Aw - 2b$$

Derivative of L_1 penalty

$$\frac{\partial}{\partial w_j} (\lambda |w_j|) = \lambda \, sign(w_j)$$

$$sign(w_j) = \begin{cases} 1 & \text{if } w_j > 0 \\ -1 & \text{if } w_j < 0 \end{cases}$$

$$\nabla_{w_j} E(w) = (2Aw - 2b)_j + \lambda \, sign(w_j)$$

The absolute value fuction $f(x) = |x|$ is coatinuous but not differentiable at $x = 0$ because the left hand and right hand limits of the derivative do not match $(+1 \neq -1)$

Therefore the gradient $\nabla_w E(w)$ does not exist at any point where at least one weight $w_j = 0$.

In practice, at $w_j = 0$, we utilize the subgradient, which is any value in the interval $[-\lambda, \lambda]$.

c)

Non-Differentiability

The L_1 penalty term has a sharp kink at $w = 0$.

Because it is not differentiable at the origin, we cannot use standart calculus to set the gradient to zero and solve directly.

Non-Linearity

The optimality condition involves the sign(w) function. This creates a non-linear equation where the solution for w depends on its own sign, making it impossible to isolate w using linear algebra.

Sparsity Selection

Lasso performs feature selection by forcing coefficients to be exactly zero. Identifying which specific weights should be zero requires an itecative search because there is no single matrix operation that can determine this sparsity pattern automatically.

d)

Given the assumption that $w_j > 0$ for all j,

$$\|w\|_1 = \sum_j w_j = 1^T w$$

$$E(w) = w^T A w - 2b^T w + c + \lambda 1^T w$$

$$\nabla_w E(w) = 2Aw - 2b + \lambda 1 = 0$$

$$2Aw^* = 2b - \lambda 1$$

$$Aw^* = b - \frac{\lambda}{2} 1$$

$$w^* = A^{-1}\left(b - \frac{\lambda}{2}1\right)$$

$$w^* = (X^TX)^{-1}\left(X^Tt - \frac{\lambda}{2}1\right)$$

---

4)

a)

$$p(t_n|x_n, w) = \frac{1}{2b}\exp\left(-\frac{|t_n - x_n^Tw|}{b}\right)$$

$$p(D|w) = \prod_{n=1}^{N} p(t_n|x_n, w) = \prod_{n=1}^{N} \frac{1}{2b}\exp\left(-\frac{|t_n - x_n^Tw|}{b}\right)$$

$$\log p(D|w) = \ln\left(\prod_{n=1}^{N} \frac{1}{2b}\exp\left(-\frac{|t_n - x_n^Tw|}{b}\right)\right)$$

$$= \sum_{n=1}^{N}\left(\ln\left(\frac{1}{2b}\right) - \frac{|t_n - x_n^Tw|}{b}\right)$$

$$= N\ln\left(\frac{1}{2b}\right) - \frac{1}{b}\sum_{n=1}^{N}|t_n - x_n^Tw|$$

b)

$$p(w_j) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{w_j^2}{2\sigma^2}\right)$$

$$p(w) = \left(\frac{1}{2\pi\sigma^2}\right)^{D/2}\exp\left(-\frac{1}{2\sigma^2}\sum_{j=1}^{D} w_j^2\right)$$

$$\log p(w) = \frac{D}{2}\ln\left(\frac{1}{2\pi\sigma^2}\right) - \frac{1}{2\sigma^2}\sum_{j=1}^{D} w_j^2$$

$$\log p(w) = \text{constant} - \frac{1}{2\sigma^2}\|w\|_2^2$$

c)

$$p(w|D) \propto p(D|w)p(w)$$

$$\log p(w|D) = \log p(D|w) + \log p(w) + \text{constant}$$

$$= \sum_{n=1}^{N}$$

$$\log p(D|w) = \text{constant}_1 - \frac{1}{b}\sum_{n=1}^{N} |t_n - x_n^T w|$$

$$\log p(w) = \text{constant}_2 - \frac{1}{2\sigma^2}\|w\|_2^2$$

$$\log p(w|D) = -\frac{1}{b}\sum_{n=1}^{N} |t_n - x_n^T w| - \frac{1}{2\sigma^2}\|w\|_2^2 + \text{constant}$$

$$\arg\max_w \log p(w|D) = \arg\min_w \left(\sum_{n=1}^{N} |t_n - x_n^T w| + \frac{b}{2\sigma^2}\|w\|_2^2\right)$$

Let $\lambda = \frac{b}{2\sigma^2}$

$$\min_w \sum_{n=1}^{N} |t_n - x_n^T w| + \lambda\|w\|_2^2$$
